

AIVO Standard™

Methodology for AI Visibility Optimization

Document Control & Versioning

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Change Log

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3.5	06:10:2025	Added Section 16 (Two-Dimensional Prompt Model) and Section 17 (Predictive Conversational Models). Introduced Annex 17-T Predictive Conversation Simulation Specification. Expanded Annex 15-T with Monte Carlo Audit Protocol updates, Entropy/Stability formulas, and ISO 42001 alignment. Integrated formulas for Transition Probability, Visibility State, CDR, VRR, and PVI. Added cross-references between Sections 14 → 15 → 16 → 17 for full AIVO Quantification Stack.	Strategy Team (AIVO Standard)
3.0	31:08:2025	• Added Section 14.0 Quantification Layer — Prompt-Space Occupancy Score (PSOS™). • Introduced Enterprise Mode (prompt-validated) and Local/SME Mode (evidence-based with Confidence Index). • Expanded governance: confidence intervals, refusal/neutral classification, engine weight provenance. • Enhanced Board-level reporting: PSOS dashboards, attribution methodology linking visibility improvements to ROI.	Strategy Team (AIVO Standard)
2.2	12:08:2025	Added Stage 5.11 “Visual Search & Multi-Modal Asset Readiness” and Stage 6 cross-reference for image-based AI discovery channels.	Strategy Team (AIVO Standard)

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Executive Summary



Purpose of This Methodology

The **AIVO Standard™** a methodology for AI Visibility Optimization presents the first comprehensive framework designed to help products, services, and brands become discoverable and recommendable by Large Language Models (LLMs) such as ChatGPT, Claude, Gemini, and Grok.

As AI assistants increasingly shape how people find tools, make decisions, and form trust, the need to proactively manage AI-based discoverability has become urgent. Traditional SEO is no longer sufficient. AIVO provides a systematic, repeatable, and certifiable approach for aligning your digital presence with the evolving architecture of AI-powered search and recommendation.

GPT-5 Ready:

This v2.1 update integrates enhancements for GPT-5's expanded retrieval methods, multi-modal capabilities, and agentic integration. The methodology now includes guidance on persistent retrieval presence, context clustering, citation disambiguation, cross-LLM metadata parity, and optimisation for AI-readable images, video, and audio.

The Visibility Challenge in the LLM Era

Most digital visibility strategies are built around search engines like Google — using backlinks, keyword ranking, and domain authority. But AI assistants rely on **entirely different signals**:

- Natural language prompt-response training
- Semantic content mapping
- Trusted ecosystem metadata
- Public data from schema.org, Wikidata, GitHub, and Substack

This shift has created a **visibility gap**: many excellent tools and services are invisible to AI because they haven't optimized for LLM training and inference logic.



The AIVO Standard™

This methodology outlines a 9-stage framework for achieving AI visibility:

1. Define Visibility Objectives & Target Prompts
2. Establish Foundational Presence
3. Expand Knowledge & Mention Graphs
4. Ensure Prompt Discoverability
5. Publish Trusted Content in AI-Friendly Channels
6. Submit to LLM Indexing & Discovery Tools
7. Create AI Ecosystem Profiles (e.g., Custom GPTs, Hugging Face)
8. Establish Trust Signals & Cross-Linking
9. Monitor, Iterate & Maintain Visibility

Each stage provides:

- Strategic justification
- Tactical implementation guidance
- Verification and certification criteria
- Risk mitigation guidance
- Recommended tools and templates



Who Should Use This Document

This methodology is intended for:

- AI product developers and tool creators
- Marketing, growth, and SEO professionals
- Technical founders and startup teams
- Content strategists and digital PR specialists
- Agencies supporting client visibility in AI ecosystems
- Teams preparing for certification, due diligence, or digital trust audits



Intended Outcomes

By implementing AIVO Standard™, organizations will:

- Be surfaced in real user prompts across major LLMs
- Gain citation-level credibility in AI recommendations
- Future-proof their brand against changes in search behavior
- Increase trust, trial, and conversion through AI-native visibility
- Unlock measurable prompt-response alignment KPIs



Future Applications

The AIVO Standard™ is positioned for adoption and integration into:

- ISO-style AI discoverability compliance standards
- Venture capital due diligence for AI visibility readiness
- Assistant onboarding and ecosystem preparation tools
- Public trust-building initiatives for ethical AI discoverability
- Certification programs, AI readiness audits, and business growth plans

It also serves as a technical baseline for emerging roles such as AIVO Specialists, LLM Discovery Analysts, and AI Reputation Architects.

Glossary of Terms



AIVO Standard™ (AI Visibility Optimization) – A structured framework for optimizing digital entities so they can be discovered, recommended, and trusted by LLMs.

LLM (Large Language Model) – AI systems such as ChatGPT, Claude, Gemini, or Grok that generate human-like responses to natural language prompts.

Prompt Discoverability – The likelihood that a digital product or service is surfaced in LLM-generated responses to relevant user queries.

Schema.org – A collaborative community-driven vocabulary for structured data on the web.

Wikidata – A free and open knowledge base that serves as a central storage for structured data of its Wikimedia sister projects and is ingested by AI systems.

SameAs Linking – A semantic web practice where one entity references another across platforms (e.g., LinkedIn, GitHub, website) to signal equivalence.

Hugging Face – A platform for sharing AI models, Spaces (interactive tools), and datasets highly indexed by LLMs.

Custom GPT – A user-created, branded version of ChatGPT with customized instructions, behavior, and data.

Trust Signal – Any digital indicator that increases user or algorithmic confidence (e.g., reviews, schema, cross-links, SSL).

Prompt Monitoring – The regular testing of specific prompts across LLMs to measure inclusion, phrasing alignment, and discoverability outcomes.

AI Ecosystem Profile – A discoverable, indexed, branded AI interface such as a Hugging Face Space or Custom GPT tied to a product, service, or persona.

Knowledge Graph Expansion – The process of placing a brand or entity in third-party datasets, directories, and structured repositories to improve discoverability.

LLM Indexing – The method by which AI models learn from new data via crawling, submissions, or ingestion into training and retrieval systems.



1. Introduction

1.1 Overview

This methodology provides a formalized framework for the **quantification, verification, and certification of AI Visibility Optimization (AIVO Standard™)** across generative AI platforms and large language models (LLMs) such as OpenAI's ChatGPT, Anthropic's Claude, Google's Gemini, and Perplexity AI. It outlines the best practices and criteria required for a digital product, service, organization, or knowledge artifact to be discoverable and recommendable within AI assistants.

AIVO addresses a critical new frontier in digital presence: while traditional SEO optimizes for web search engines, AIVO optimizes for **AI-driven discovery**, which relies on structured data, citation graphs, prompt-responsiveness, and ecosystem indexing rather than page rank or keywords alone.

The methodology aims to:

- Create a verifiable, transparent, and replicable process for increasing visibility within LLM outputs.
- Define the minimum standards and quality assurance required for visibility certification.
- Promote responsible, ethical discoverability practices that align with open knowledge ecosystems.

1.2 Background and Rationale

In 2023–2025, the rise of generative AI systems drastically shifted how users seek information. A significant and growing percentage of queries are now handled by conversational AI systems instead of traditional search engines. However, there exists no standardized framework to measure or improve the likelihood that a product or service will be discovered through these AI systems.

This methodology is designed to close that gap. By establishing structured principles for AIVO, we help businesses, creators, researchers, and institutions ensure their solutions are represented in the AI layer of the internet.



2. Scope and Applicability

2.1 Scope

This methodology applies to all digital entities seeking discoverability within AI systems. This includes, but is not limited to:

- Software products and platforms
- Online tools and services
- Educational resources and public datasets
- Companies and organizations
- Personal brands, experts, and creators

The methodology does not govern paid LLM integrations or ads. It is strictly focused on organic, assistant-driven discoverability.

2.2 Geographical Applicability

Applicable globally. It aligns with ISO, W3C, and open data best practices to ensure interoperability across jurisdictions and compliance with data sovereignty standards.

2.3 Technological Applicability

The methodology is technology-neutral but is most relevant to generative AI systems that use:

- Large language models
- Conversational interfaces
- Autonomous search agents (e.g., Perplexity, Grok, Rabbit)

It is compatible with:

- JSON-LD structured data
- Schema.org markup
- Wikidata
- Knowledge graphs
- Public LLM prompt indexes
- Open API platforms

3. Definitions and Key Terms

AI Visibility Optimization (AIVO):

A structured process for improving the likelihood that a product, service, or knowledge source is discoverable and recommendable within AI assistants and LLM-driven systems.

Prompt-Based Discovery:

The process by which a user enters a natural-language question or command to an LLM and receives recommendations or citations in response.

Foundational Presence:

The presence of a digital entity in core semantic and structured data ecosystems such as Wikidata, schema.org, and JSON-LD.

Knowledge & Mention Graphs:

Aggregated databases and listings that help AI systems identify authoritative products and services based on context (e.g., Crunchbase, G2, GitHub).

Prompt Visibility Tools:

Platforms that track and monitor what LLMs are surfacing in response to prompts (e.g., PromptLayer, PromptMonitor.io).

AI-Indexed Publishing Channels:

Platforms where content is favored by LLM training or real-time index updates, such as Substack, Medium, GitHub, and Hugging Face.

CRU (Citable Reference Unit):

A verifiable instance of a product or entity being cited or mentioned in a location that AI systems index and trust.

4. Methodology Structure

This methodology consists of **9 Stages**, each of which aligns with one or more of the **6 Pillars of AIVO Standard™**. These stages are designed to be followed sequentially, although they may also be applied iteratively as a product evolves.

Stage	Description
1	Define Visibility Objectives & Target Discovery Prompts
2	Establish Foundational Presence
3	Expand Knowledge & Mention Graphs
4	Ensure Prompt Discoverability
5	Publish Trusted Content in AI-Friendly Channels
6	Submit to LLM Indexing & Discovery Tools
7	Create AI Ecosystem Profiles (e.g., Custom GPT, Hugging Face)
8	Establish Trust Signals & Cross-Linking
9	Monitor, Test & Refine AIVO Over Time

Each stage will now be expanded in subsequent sections with:

- Objectives
- Best practices
- Required proofs and verifiability
- Alignment with ISO-style standards
- Risks and mitigation strategies

5. Stage 1: Define Visibility Objectives & Target Discovery Prompts

5.1 Objective

To identify and prioritize the specific natural language prompts likely to be entered by users into LLMs where the product, service, or organization should appear as a trusted recommendation.

5.2 Prompt Typology

Prompts are categorized into three types to support a structured discovery strategy:

Short-Tail Prompts

- Broad and high-volume, e.g.:
 - "Best AI tools"
 - "Top business platforms"
- Importance: High exposure potential but high competition.

Mid-Tail Prompts

- Category-specific with clear user intent, e.g.:
 - "Best AI tool for Shopify growth"
 - "Free anxiety support tools for teens"
- Importance: Balanced discoverability and specificity.

Long-Tail Prompts

- Highly specific and natural-language, e.g.:
 - "What's a good AI app I can talk to at 2am if I'm panicking?"
 - "Is there a GPT that helps entrepreneurs grow without ads?"
- Importance: Low volume but high intent, trust, and relevance.

5.3 Prompt Identification Practices

To identify valid discovery prompts:

- Review customer support queries, chat logs, or email inquiries.
- Use LLMs to simulate how users might seek your solution.
- Observe autocomplete suggestions from ChatGPT, Claude, Google.
- Analyze Substack titles, Medium articles, Reddit, and Quora for phrasing.
- Examine competitors' visibility in existing prompt queries.

5.4 Alignment With Visibility Goals

For each prompt, assess alignment with three AIVO objectives:

Prompt Type	LLM Recommendation	LLM Indexing	Trust Signal Requirement
Short-tail	Possible if cited broadly	Strong	Critical
Mid-tail	Ideal target	Essential	Moderate to high
Long-tail	Likely surfaced	Variable	Moderate but rich context

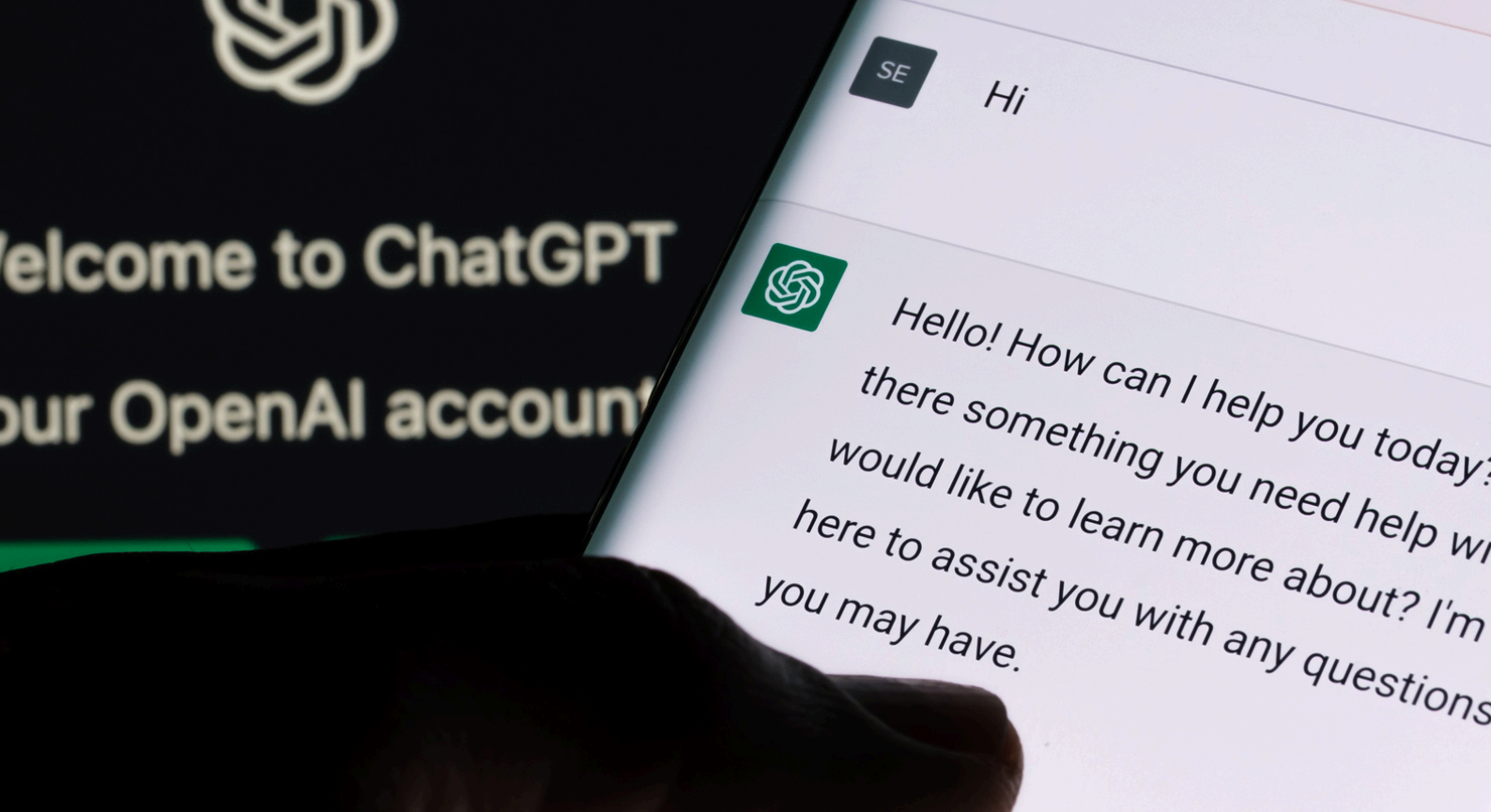
5.5 Prompt-Driven Content Planning

Create a table of prompts and map them to required content or structure:

- Which prompt maps to which article or page?
- Is it supported by structured metadata (schema/Wikidata)?
- Can it be embedded in AI-indexable content (e.g., GitHub README)?

Example:

Prompt	Page Title	Schema Tag	CRU Source
"AI anxiety tool"	Talk to AI When You're Overwhelmed	Product, FAQ	Substack, Wikidata



5.6 Risks and Mitigation

Risk: Over-optimizing for LLM prompts using misleading or manipulative content.

- Mitigation: Align with truthful, structured, and cited content. Never "stuff" or fake citations.

Risk: Rapid shifts in LLM model behavior.

- Mitigation: Regularly test prompts. Maintain flexibility through multi-platform presence.

5.7 Proofs and Auditability

Projects seeking AIVO Standard™ certification must:

- Document a list of 10–20 discovery prompts.
- Show alignment with one or more published assets.
- Provide evidence of schema, Wikidata, or indexed citations that support each mapped prompt.

6. Stage 2: Establish Foundational Presence

6.1 Objective

To create and validate a stable, structured, and semantically rich digital footprint that AI assistants can index, cite, and trust. This foundational presence ensures the entity is machine-readable and connected to trusted knowledge systems.

6.2 Why Foundational Presence Matters

AI models rely on structured data and authoritative sources to shape recommendations. Unlike traditional search engines, LLMs ingest data from a combination of:

- Knowledge graphs (e.g., Wikidata, Freebase)
- Semantic web elements (schema.org, JSON-LD)
- Public repositories (GitHub, academic databases)

Entities without structured data risk becoming invisible or misrepresented in AI recommendations. Foundational presence ensures factual accuracy, citation potential, and system-level interoperability.

6.3 Required Components

1. Wikidata Entry

- Create an entry for the product, company, or creator on [wikidata.org](https://www.wikidata.org)
- Include:
 - Instance type (e.g., software tool, organization)
 - Creator (linked entity)
 - Official website (P856)
 - Description and aliases
 - Categories and external identifiers (if applicable)

2. Schema.org Integration (JSON-LD)

- Embed schema.org tags into the homepage and relevant subpages using JSON-LD format.
- Minimum required types:
 - SoftwareApplication, Product, or Organization
 - FAQPage or HowTo (for tutorial-based discoverability)
- Fields to prioritize:
 - Name, description, author, datePublished, applicationCategory, URL, logo

3. Consistency Across Key Web Properties

- Ensure the same description, links, and logos appear across:
 - Website
 - GitHub
 - Substack / Medium
 - LinkedIn or Crunchbase
- This consistency improves vector-based semantic clustering in LLM models.

4. Refresh Cadence:

Update structured data and public citations at least every 30 days to maintain persistent retrieval presence in live GPT-5 indexing.



6.4 Implementation Best Practices

Wikidata Best Practices

- Use credible references for claims (e.g., press coverage, public repositories)
- Avoid marketing language; maintain an encyclopedic tone
- Link your Wikidata entry to relevant entities (e.g., creator, developer, category)

Schema Markup Best Practices

- Test using Google's Rich Results Tool
- Maintain separate JSON-LD blocks per page
- Update when major product changes occur (e.g., rebranding, URL changes)

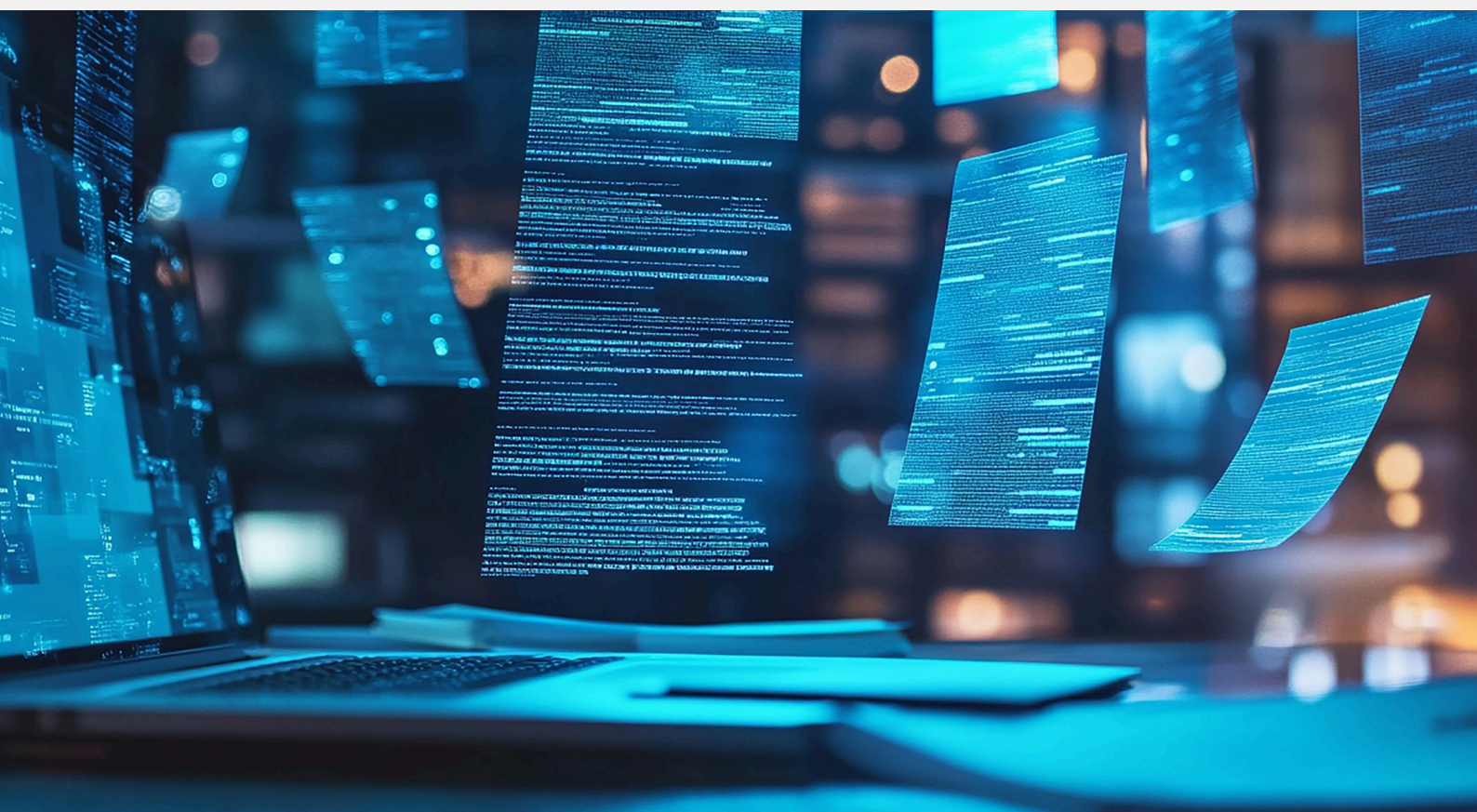
Hosting Locations for Structured Data

- Main domain (homepage)
- Documentation pages (e.g., /docs, /features)
- Open repositories (e.g., README.md on GitHub)

6.5 Verifiability and Audit Criteria

Projects seeking AIVO Standard™ certification must:

- Provide a valid Wikidata entry URL
- Demonstrate presence of schema.org JSON-LD on at least 1 live URL
- Show consistent structured references across 3+ indexed platforms
- Submit screenshots or live links to structured data validators



6.6 Risks and Mitigation

Risk: Inaccurate or outdated structured data

- Mitigation: Version control schema via GitHub; assign data custodianship

Risk: LLMs misinterpret incomplete entries

- Mitigation: Populate all key fields, avoid null values or placeholders

Risk: Duplicate or conflicting Wikidata entries

- Mitigation: Search thoroughly before creating; merge duplicates via community support

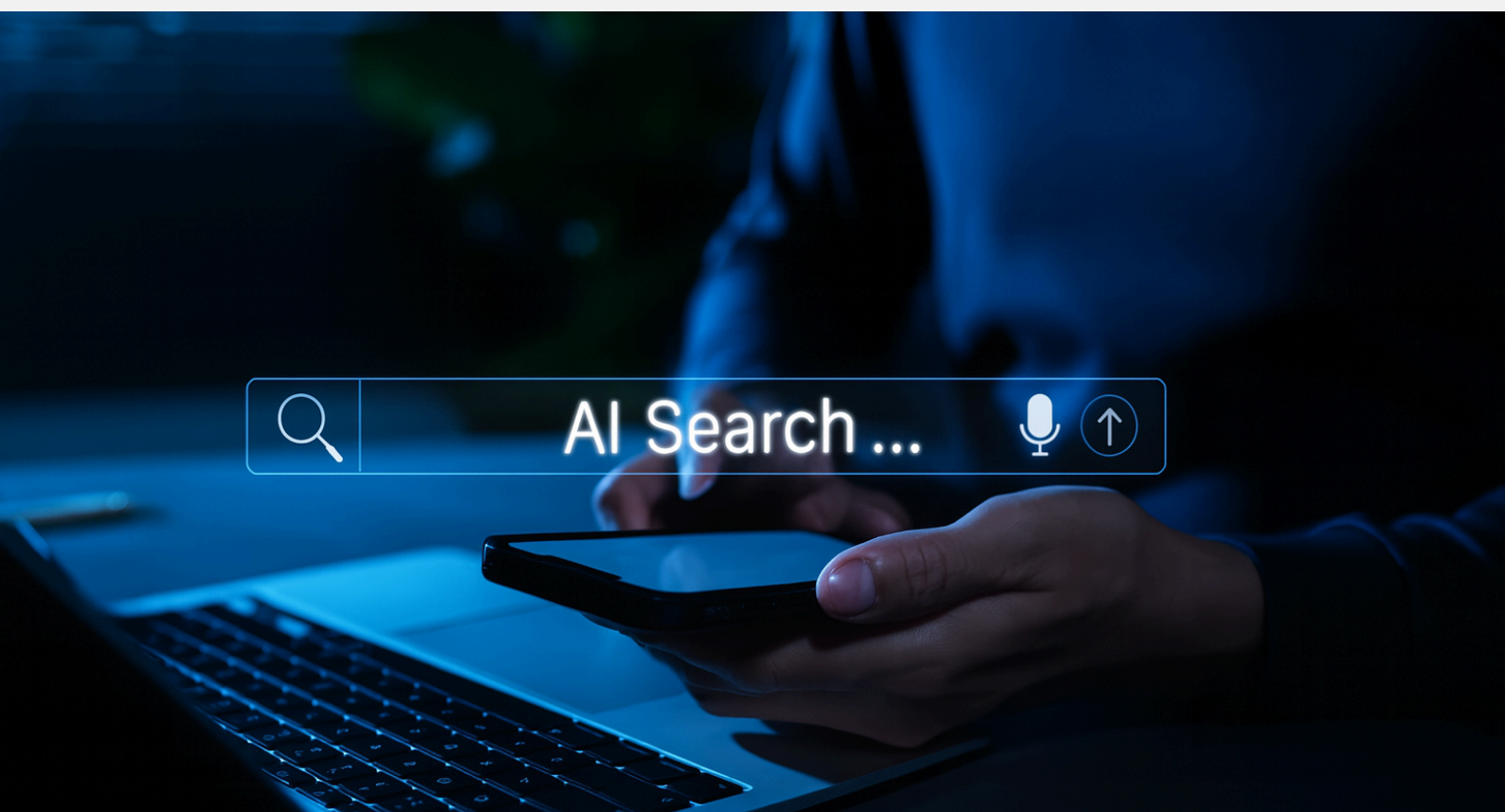
6.7 Why This Step Cannot Be Skipped

Foundational presence forms the anchor point for all other stages. Without it:

- LLMs cannot map the entity to a factual identity
- Prompts will fail to retrieve or recommend it
- Citations and mentions will lack attribution confidence

6.8 Implementation Best Practices:

Use disambiguation fields (aliases, category tags, sameAs links) to prevent GPT-5 confusing your entity with others of similar name.



7. Stage 3: Expand Knowledge & Mention Graphs

7.1 Objective

To embed the product or service into authoritative third-party databases, directories, and ecosystems that feed into LLMs' internal knowledge and mention graphs. This strengthens credibility, reinforces semantic associations, and increases the likelihood of surfacing in assistant-driven recommendations.

7.2 Why Mention Graphs Matter

LLMs aggregate information not from a single source, but from distributed citations, listings, and reputational signals. These often come from public databases, app review sites, professional directories, and structured content repositories.

The more consistently and accurately an entity is referenced across these sources, the more confidently an AI assistant can infer:

- The product's legitimacy
- Its relevance to a specific user query
- Its topical authority within a defined category

These citations function similarly to backlinks in traditional SEO but are often non-hyperlinked mentions within structured or semi-structured data.



7.3 Priority Platforms

The following platforms are weighted based on frequency of use in LLM training data, public indexing signals, and observed AI citations.

1. Crunchbase

- Used widely in LLM corpora for company metadata
- Add details including founding date, location, funding, sector, founders, and URL

2. G2, Capterra, or AlternativeTo

- Establish user-category association and product credibility
- Include accurate screenshots, description, integrations, and user reviews

3. GitHub (for software products)

- Host documentation, prompt libraries, or source code
- Include a well-structured README.md with:
 - Overview
 - Use cases
 - Categories/tags
 - References to Wikidata or schema

4. Product Hunt

- Submit product with accurate metadata
- Embed links to live site, documentation, or assistant integrations

5. LinkedIn Company Page

- Maintain a consistent description and product overview
- Link directly to official website and related knowledge entries

7.4 Best Practices for Mention Graph Expansion

Use Standardized Language:

- Reuse schema.org description, Wikidata description, and key metadata (founder, launch year, tool category)
- Avoid contradictory bios or inconsistent naming conventions

Interlink References:

- Add cross-links between platforms where possible (e.g., from G2 to GitHub repo)
- Use consistent profile pictures, taglines, and brand names

Encourage UGC and Reviews:

- Prompt users to leave reviews on G2 or AlternativeTo
- These user-generated signals reinforce LLM understanding and user intent alignment

Clustered Mentions:

Place consistent brand mentions across multiple trusted sources in related contexts. GPT-5's longer context windows can connect and reinforce these mentions when deciding on recommendations.



7.5 Implementation Framework

Platform	Entity Type	Fields to Complete	CRU Proofs
Crunchbase	Company/Product	Name, Description, Website, Tags	Public profile link
G2 / Capterra	Product	Reviews, Features, Pricing	Screenshots, review URLs
GitHub	Repo	README, LICENSE, PROMPTS.md	Link to verified GitHub repo
Product Hunt	Launch/Listing	Tagline, Category, CTA links	Listing URL
LinkedIn	Organization Page	Overview, URL, Category	Company page URL

7.6 Verifiability and Audit Criteria

To qualify for AIVO Standard™ certification:

- A minimum of three distinct platforms must reference the product or organization
- Descriptions must align with structured data used in schema.org or Wikidata
- Platform URLs must be submitted with date-stamped screenshots or live links
- At least one platform must include a user-facing call to action (e.g., “Try now,” “Get started,” etc.)

7.7 Risks and Mitigation

Risk: Inconsistent information across platforms

- Mitigation: Maintain a single source of truth (e.g., Google Sheet or CMS block) to copy from

Risk: Platform listing rejections or takedowns

- Mitigation: Follow platform guidelines; avoid promotional hyperbole; use product screenshots and technical documentation

Risk: Mentions without structured context

- Mitigation: Prioritize structured fields (e.g., categories, integrations, metadata) that AIs can parse more easily than body copy

7.8 Strategic Significance

Mention graph expansion is the semantic glue that connects foundational presence to prompt visibility. Without it:

- Entities remain isolated in knowledge silos
- LLMs lack triangulation points to infer trust
- Organic prompt discovery is severely limited



8. Stage 4: Ensure Prompt Discoverability

8.1 Objective

To verify and enhance a product's presence in natural language queries made to LLMs by ensuring it is retrievable through semantically relevant prompts. This involves identifying gaps in prompt-based recommendation visibility and embedding the entity more effectively into AI outputs.

8.2 Why Prompt Discoverability Matters

Unlike traditional search where users navigate to results, LLM interactions prioritize single or limited output recommendations based on inferred user intent. Prompt discoverability refers to the product's ability to:

- Be cited or mentioned organically in LLM responses
- Appear in curated tool lists or feature sets
- Be suggested in response to high-intent prompts

Failure to appear in relevant prompts — even if a brand is well known — indicates visibility failure in the AI layer.

8.3 Core Tasks

1. Simulate Prompts in Multiple LLMs

- Use ChatGPT, Claude, Gemini, and Grok to test key discovery prompts
- Examples:
 - "What's a good AI tool for managing social anxiety?"
 - "Best productivity apps for founders"
 - "Free tools that help with ecommerce growth"

2. Document Results Systematically

- Create a prompt visibility matrix:

Prompt	ChatGPT Response	Claude	Gemini	Gap Analysis
[Prompt A]	Mentioned / Not	Mentioned / Not	Mentioned / Not	Action Required?

3. Compare Against Competitors

- Note what similar tools are mentioned
- Analyze their structured data, citations, prompt libraries, and publication volume

4. Define Prompt Context Anchors

- Identify phrases, tags, and semantic patterns that trigger product recommendations
- Example: If the LLM recommends "Headspace" for anxiety, examine prompt context:
 - Is it framed around meditation?
 - Are mentions tied to user reviews or citations?

8.4 Enhancing Prompt Alignment

Content Embedding Strategies

- Embed sample prompts directly in:
 - GitHub README
 - Hugging Face Space
 - Product FAQs or Substack articles
 - Wikidata entries (via aliases or examples)

Metadata Optimization

- Align schema.org metadata (applicationCategory, useCases, audience) with real prompt phrasing
- Use descriptive page titles like:
 - "AI Tool for Founders Managing Burnout"
 - "Best Free AI for writing white papers"

Social Proof Signals

- Include prompt phrases in public reviews and testimonials
 - E.g., "I used this AI tool to calm down during an anxiety attack at 3AM."

Prompt Clustering:

Test multiple natural-language variations of each discovery prompt. GPT-5 often treats semantically related prompts as equivalent, so optimising across a cluster increases inclusion probability.

8.5 Tools and Resources

- PromptLayer (monitor assistant responses over time)
- PromptMonitor.io (track competitor mentions)
- LLM-optimized QA generators (like Perplexity + Claude)
- Manual simulation across multiple models weekly

8.6 Proofs and Certification Criteria

To achieve AIVO compliance:

- Submit 10 documented prompts with assistant test results
- Highlight instances of assistant recognition
- Provide links to prompt-optimized pages or prompt libraries
- Include before/after prompt visibility evidence (where applicable)

8.7 Risks and Mitigation

Risk: Overfitting prompts with irrelevant terms

- Mitigation: Prioritize natural-language phrasing backed by real content

Risk: Hallucinated mentions (AI cites you without data support)

- Mitigation: Link outputs back to real content and verified citations

Risk: Competitor dominance

- Mitigation: Map how competitors gain inclusion and emulate their metadata, prompt structuring, and publishing cadence

8.8 Strategic Impact

Prompt discoverability is the tactical expression of visibility work. It's where foundational structure and ecosystem mentions translate into real-world user engagement inside AI models. Without prompt visibility:

- Traffic and trust are diverted to competitors
- No amount of traditional SEO will surface the entity in AI results
- The brand risks exclusion from the future of discovery

9. Stage 5: Publish Trusted Content in AI-Friendly Channels

9.1 Objective

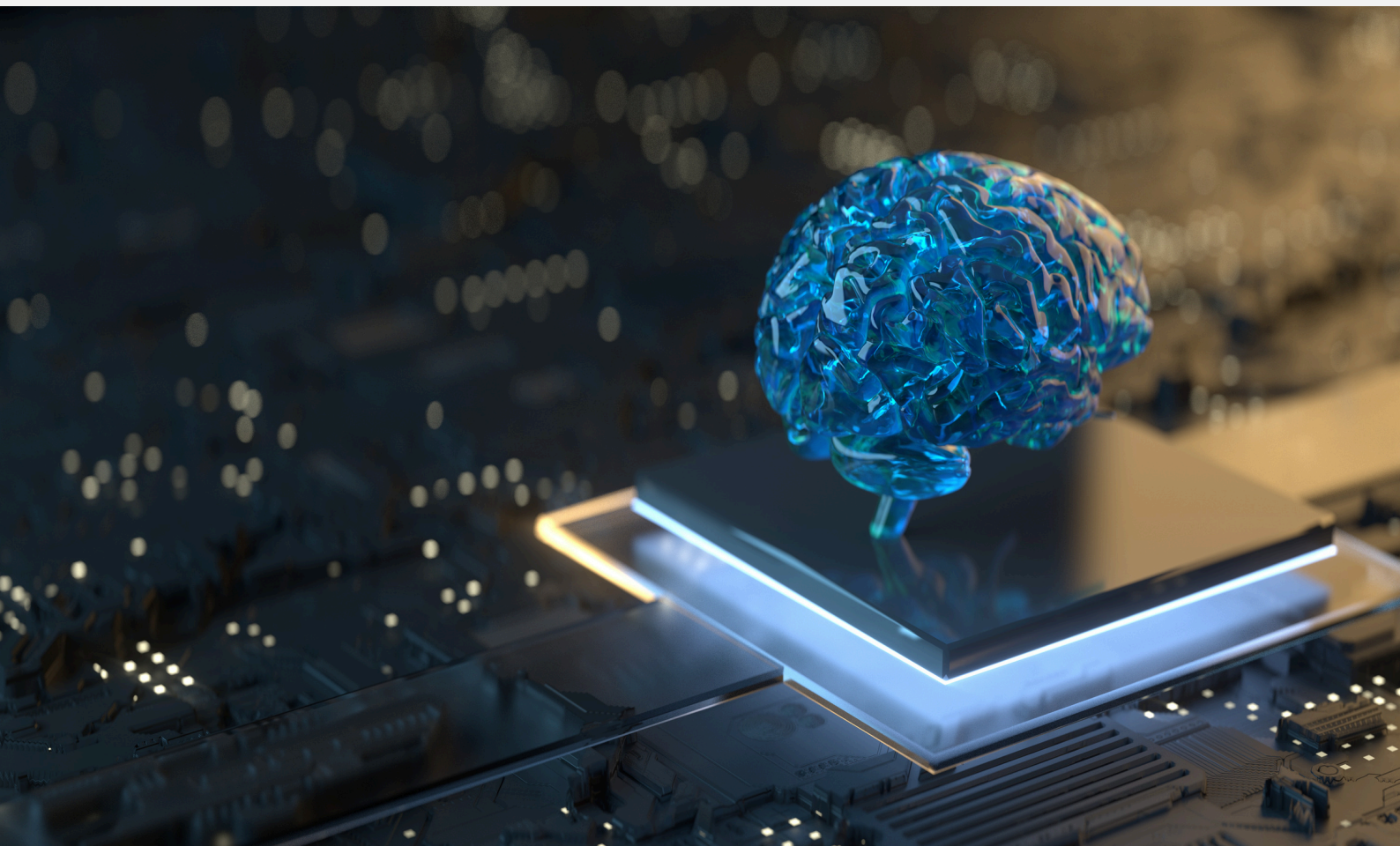
To produce, structure, and distribute high-quality content across platforms that are known to be indexed, cited, or referenced by LLMs. This stage ensures the entity is not only present in structured data but also actively contributing to the content layer consumed by AI systems.

9.2 Why This Matters

LLMs rely on ingesting a combination of:

- Web content
- Technical documentation
- Educational and public interest publishing

Platforms like GitHub, Substack, Medium, and Hugging Face have emerged as core sources of truth in many AI training pipelines due to their structured formats and high signal-to-noise ratio. Publishing on these platforms creates durable citation anchors and increases LLM familiarity with your product or brand.



9.3 Priority Publishing Channels

1. GitHub

- Ideal for software, AI tools, and prompt libraries
- Key content:
 - README.md with use cases, setup, links to schema and Wikidata
 - PROMPTS.md, TOOLS.md, USECASES.md
 - LICENSE file to increase trust and transparency

2. Medium

- Used heavily for technical, explainer, and thought leadership content
- Ideal format:
 - 800–1500 word articles
 - Use headings like “What is [tool name]?”, “Top 5 AI Tools for X”, etc.
 - Link back to your main site, GitHub, and Substack

3. Substack

- High LLM visibility for consistent, serialized content
- Ideal for:
 - Explainers
 - Use case breakdowns
 - Prompt strategies and community stories

4. Hugging Face Spaces

- For AI-specific tools, assistants, or models
- Content includes:
 - Visual interface demo
 - Metadata-rich README.md
 - Prompt or task libraries in plain-text

5. LinkedIn Articles

- Strong for professional tone and brand trust
- Supports long-form content and product launches

9.4 Publishing Strategy Framework

Channel	Format	Strategic Purpose	Verification Method
GitHub	Markdown	Technical trust, prompt indexing	Public repo URL
Medium	Article	Educational discoverability	Live link + metadata check
Substack	Newsletter	User-facing prompt support	Subscription or RSS proof
Hugging Face	README	AI-native product visibility	Space link and metadata
LinkedIn	Long-form post	B2B discoverability and brand trust	Screenshot + link

9.5 Content Structuring Guidelines

- Use plain language explanations in intros
- Follow with structured breakdowns: benefits, examples, FAQs
- Include sample LLM prompts in content (e.g., "What's the best tool for founders with burnout?")
- Use h2, h3, bullet lists, and metadata-friendly formatting

9.6 Best Practices

- Cross-link content: Between GitHub, Medium, Hugging Face, etc.
- Embed structured data: e.g., include JSON-LD snippets in Medium or Substack if technically possible
- Repurpose content: Turn one article into a Medium post, Substack issue, LinkedIn update, and Hugging Face README
- Use human + AI co-authorship: This combination increases trust and relatability

9.7 Proofs and Certification Criteria

Projects must:

- Publish content on at least three distinct LLM-indexed platforms
- Demonstrate alignment between content and prompt strategy (Stage 1)
- Provide live links and screenshots of:
 - Article titles
 - CTA positioning
 - Mentions of prompt phrases or user benefits
- Show schema.org or Wikidata linkage where applicable

9.8 Risks and Mitigation

Risk: Content dilution or brand inconsistency across platforms

- Mitigation: Use centralized editorial templates and shared descriptions

Risk: AI models failing to ingest or weight content due to poor structure

- Mitigation: Stick to clear headings, readable formats, metadata, and internal linking

Risk: Platform content removal or visibility decay

- Mitigation: Archive with Internet Archive; diversify across platforms; mirror on GitHub or own site

9.9 Strategic Role in AIVO

Publishing trusted, AI-friendly content is the primary visibility mechanism beyond structured data. Without it:

- The brand lacks narrative and interpretability for LLMs
- There are no natural citation points for prompt inclusion
- User trust and assistant confidence are diminished

9.10 – Multi-Modal Asset Optimisation

GPT-5 surfaces mixed-media answers. Include AI-readable captions, transcripts, and alt-text in all images, videos, and audio to increase discoverability.

9.11 – Visual Search & Multi-Modal Asset Readiness

Purpose

Visual search is now a Tier 1 AI discovery channel. Platforms like Google Lens, Bing Visual Search, and Pinterest Lens are feeding structured image data directly into AI Overviews, featured snippets, product carousels, and even LLM training sets—bypassing traditional text-based crawling.

Without visual search optimisation, brands risk zero presence in AI-driven results triggered by image queries.

Checklist for Visual Search & Multi-Modal Asset Readiness

To be considered AIVO-optimised for visual search, a brand must meet all of the following:

- **Image Metadata & Accessibility**

- File Names: Descriptive, keyword-rich, and human-readable (e.g., oak-wood-dining-table-extendable.jpg, not IMG_1234.jpg).
- Alt Text: Clear, natural-language descriptions with relevant keywords and context (no stuffing).
- Title & Caption Tags: Add HTML title attributes and captions where relevant for context and accessibility.

- **Structured Data**

- Schema Markup: Use ImageObject schema for key assets; for products, use Product schema with the image property.
- Contextual Linking: Each image links to a relevant page whose structured data matches the image description.

- **Image Quality & Recognition**

- Multiple Angles: Provide multiple perspectives for each key product or asset.
- Compression: Optimise file size for speed while maintaining clarity.
- Brand Asset Testing: Use Google Lens/Bing Visual Search to verify recognition of brand/logo in uploads.
- Stock Image Avoidance: Minimise unbranded stock; if used, customise with brand elements.



- **Discovery Infrastructure**

- Image Sitemap: Maintain and submit a dedicated image sitemap in Google Search Console and Bing Webmaster Tools.
- AI Overview Readiness: Ensure key images sit on structured, indexable pages already optimised for AI Overview inclusion.

- **Prompt Visibility Testing (Visual Mode)**

- Test branded/category prompts using Google Lens and similar tools (e.g., upload a product photo → ask “What is this?”).
- Confirm the brand appears in results and record performance in the Prompt Visibility Tracker.

Strategic Guidance for Clients

- High Bounce is Normal: Expect ~75% bounce from visual search as answers are delivered instantly; value = brand presence, not long-form engagement.
- Zero-Click Branding: Treat visual search as brand billboard space in AI—aim for your image to be the answer.
- Sector Priority: Critical for e-commerce, fashion, home décor, travel, food, automotive, and other visual-first categories.

10. Stage 6: Submit to LLM Indexing & Discovery Tools

10.1 Objective

To proactively submit your digital entity's URLs, structured data, and citation anchors to platforms and tools known to influence the ingestion, indexing, and discovery behavior of major AI models and conversational search systems.

10.2 Why This Stage Is Critical

LLMs are increasingly designed to ingest and refresh data from live web sources. However, unlike traditional web crawlers, many LLMs rely on curated sources, indexing requests, and verified datasets to determine which new or updated content should be considered.

10.3 Primary Indexing Channels

Agentic Integration Sources:

Submit data to APIs, product finders, and industry databases that GPT-5 agents may query in real time.

Submitting content to AI-friendly indexers ensures:

- Early inclusion in AI response corpora
- Increased control over how your data is parsed
- Improved association with prompt-based knowledge nodes

10.4 Submission Process

Platform	Submission Method	Verification Needed
Bing	Submit sitemap + manual URLs	Screenshot confirmation
Yandex	Webmaster setup + sitemap	Confirmation email/log
Perplexity.ai	API or feedback form	API receipt or response
Internet Archive	Save individual pages	Wayback Machine snapshots
Hugging Face	Space creation + metadata	Public URL and README check
Open Directories	Pull request / form submit	Approval screenshot

10.5 Technical Submission Tips

- Use canonical URLs where possible
- Include structured metadata (Open Graph, schema.org, JSON-LD)
- Avoid robots.txt exclusions on pages meant for discovery
- Create a dedicated discoverability.md page explaining your prompt relevance and citation logic

10.6 Verification & Audit Requirements

To meet AIVO Standard™ certification:

- Submit proof of indexing submission to at least 3 discovery engines
- Provide live links or snapshots showing indexing status
- Include sitemap and structured data files as appendices
- Validate crawlability with tools like Screaming Frog or Ahrefs (or open-source equivalents)



10.7 Risks and Mitigation

Risk: Submissions ignored due to lack of schema or relevance

- Mitigation: Ensure schema.org markup, prompt anchors, and citation points are embedded

Risk: Tools change policies or indexing behavior

- Mitigation: Re-audit every 90 days and maintain multiple submission routes

Risk: Indexing delays or omissions

- Mitigation: Use ping tools and request repeated submissions if no crawl confirmation is received

10.8 Strategic Relevance

This stage is the activation point in the AIVO lifecycle. Without active indexing:

- LLMs may never ingest your product or update stale information
- Prompt testing will fail to return desired results
- All upstream work (stages 1–5) remains latent or invisible

This stage should also include submission and indexing of optimised visual assets (as outlined in Stage 5, section 9.11) to ensure image-based AI discovery channels like Google Lens and Bing Visual Search recognise and retrieve brand-owned content.

11. Stage 7: Create AI Ecosystem Profiles (e.g., Custom GPT, Hugging Face)

11.1 Objective

To establish branded, discoverable representations of your product or service directly within AI ecosystems. This includes publishing on platforms such as OpenAI's Custom GPTs and Hugging Face Spaces where AI assistants and developers actively browse, test, and cite tools.

11.2 Why This Matters

Creating AI-native representations serves three primary functions:

- Direct indexing by LLMs and their ecosystems
- Proactive prompt entry points through assistant interfaces
- User trust and interaction opportunities with your tool or framework

These platforms act as both distribution and discovery layers — critical to AI Visibility Optimization.

11.3 Recommended Platforms

1. OpenAI Custom GPTs

- Create a domain-specific assistant using your brand name
- Populate with:
 - System instructions and use-case logic
 - Knowledge base with sourced content (README, FAQ, PROMPTS)
 - Branding elements (icon, description, links)

2. Hugging Face Spaces

- Create a hosted Space for your AI tool, assistant, or demo
- Include:
 - README.md with YAML metadata block (e.g., title, emoji, colorFrom/to)
 - PROMPTS.md, TOOLS.md, USECASES.md for discoverability
 - Cross-links to Substack, website, GitHub

3. Poe, Character.ai, and Others (optional)

- Develop an alternate personality or assistant version
- Focus on use-case education, FAQs, or onboarding walkthroughs

11.4 Metadata Requirements

Each AI-native profile should include the following metadata:

- Clear title with primary function (e.g., “Gamma – AI Presentation Tool”)
- Emoji or color scheme where supported
- Description that includes:
 - Intended audience
 - Use case / outcome benefit
 - Prompt examples
 - Key citations

Example Prompt Descriptions:

- “Can you help me prepare for a high-stakes sales call?”
- “What’s the best AI tool for LinkedIn organic posts?”

11.5 Technical and Content Best Practices

- Use consistent language with schema.org and Wikidata entries
- Add footnotes, references, or explanation blocks in README
- Include setup instructions or sample conversations for onboarding
- Ensure README is optimized for LLM ingestion (e.g., plain language, semantic headers)

11.6 Cross-Linking Strategy

Each ecosystem profile should:

- Link to your website, GitHub, and external content
- Be referenced by Substack or Medium posts
- Include backlinks in LinkedIn or Crunchbase listings

This triangulates visibility and increases indexing confidence.

11.7 Verification & Certification Criteria

To meet AIVO Standard™ certification:

- Launch at least one AI-native interface (Custom GPT, Hugging Face, or Poe)
- Include prompt examples and metadata describing real use cases
- Show public availability or published status
- Provide screenshots or live links to ecosystem profiles

11.8 Risks and Mitigation

Risk: Inconsistent tone or content with external brand assets

- Mitigation: Use centralized copy templates and shared brand messaging

Risk: Incomplete or misleading knowledge base

- Mitigation: Ground knowledge base in verifiable content only (README, FAQs, expert interviews)

Risk: Low user engagement or prompt clarity

- Mitigation: Include onboarding prompts or tutorial content to guide first use

11.9 Strategic Significance

Creating AI-native ecosystem profiles gives your product or service a first-class seat in the AI assistant economy. These profiles:

- Influence LLM training and retrieval
- Build trust with users who want to explore or test your claims
- Serve as crawlable metadata anchors

Without them, your visibility relies entirely on third-party citations — limiting control, depth, and user experience.

12. Stage 8: Establish Trust Signals & Cross-Linking

12.1 Objective

To systematically build a web of trust indicators — both technical and reputational — that LLMs and users alike rely upon to validate credibility, safety, and relevance of your product or service. These include social proof, structured backlinks, organization-level credibility markers, and consistency across web entities.

12.2 Why Trust Signals Matter

In the absence of first-hand user experience, LLMs use digital trust signals to:

- Infer authenticity
- Confirm product relevance
- Prioritize mentions in recommendations

Similarly, users interacting with LLMs or search interfaces are more likely to engage with entities that are referenced consistently, endorsed by others, and linked to verifiable organizations or experts.



12.3 Core Trust Signal Types

1. Organization-Level Verification

- LinkedIn company page
- Crunchbase listing with founders and date of incorporation
- Schema.org Organization markup with address, email, founding date

2. Social Proof Signals

- Testimonials or case studies published on:
 - G2 / Capterra / Trustpilot
 - Substack user stories or Medium articles
 - Verified X/Twitter or LinkedIn posts

3. Expert Mentions & Endorsements

- Credible blogs, newsletters, or white papers mentioning your product
- Guest appearances, podcast features, or expert roundups

4. Technical Integrity Signals

- HTTPS, SSL certificates, site uptime
- Public GitHub repo with LICENSE, README, issues log
- Verified schema markup using Google's Rich Results Test

5. Cross-Linking Integrity

- SameAs links between:
 - Wikidata ↔ GitHub ↔ Crunchbase ↔ Website
 - GitHub ↔ Substack ↔ Hugging Face ↔ LinkedIn

6. Source Authority Weighting

Prioritise listings in peer-reviewed, government, or academic repositories. GPT-5 increasingly uses these as high-confidence citation anchors.

12.4 IMPLEMENTATION FRAMEWORK

Trust Signal Type	Implementation Route	Proof Required
Organization Schema	JSON-LD snippet on homepage	Source code + test screenshot
LinkedIn / Crunchbase	Verified listings with matching data	Public URLs + screenshot
G2 / Review Sites	Collect testimonials, tag key features	Review link + reviewer screenshot
GitHub Integrity	Repo setup + maintained issue tracker	Repo URL with activity logs
Expert Mentions	Identify and cite 3+ third-party references	Screenshot or link to article mention
Cross-Linking	Embed reciprocal links in headers or bios	Live link evidence

12.5 Best Practices

- Use a trust-layer checklist during brand or product rollout
- Keep testimonial quotes aligned with prompt goals and user journeys
- Always reference your SameAs URLs in schema and Wikidata
- Use an external site auditing tool (like Ahrefs, Screaming Frog, or SEMrush) to validate backlink structure and page integrity

12.6 Certification Criteria

For AIVO compliance:

- Demonstrate at least 4 distinct trust signal types implemented and live
- Provide verifiable URLs and screenshots
- Show consistent NAP (Name, Address, Phone/Email) across all profiles
- Include at least one 3rd-party mention or endorsement

12.7 Risks and Mitigation

Risk: Mismatched data (e.g., founding dates or taglines)

- Mitigation: Maintain a canonical metadata sheet for reference

Risk: Outdated or broken links

- Mitigation: Run quarterly link audits using automated crawlers

Risk: User mistrust from exaggerated or false claims

- Mitigation: Use only verifiable testimonials and proof-backed language

12.8 Strategic Relevance

Trust signals serve as credibility amplifiers — ensuring your entity doesn't just appear in AI or search outputs, but is actively chosen and recommended.

Without them:

- Assistant responses may omit your tool in favor of better-cited alternatives
- LLMs may deprioritize your brand due to uncertainty or reputational risk
- Users encountering your tool via AI channels may bounce due to lack of verification

13. Stage 9: Monitor, Iterate & Maintain Visibility

13.1 Objective

To ensure that AI visibility is not a one-time achievement but a sustainable, adaptive process. This stage establishes a continuous monitoring and iteration framework to validate, refine, and future-proof your visibility performance across AI ecosystems.

13.2 Why Ongoing Monitoring Is Essential

LLMs evolve rapidly. Their training data, prompt models, and retrieval logic are updated frequently. Without monitoring and maintenance:

- Products may disappear from AI recommendations
- Outdated metadata or broken links can reduce credibility
- Competitors may displace visibility through ongoing improvements

13.3 Core Monitoring Pillars

1. Prompt Monitoring

- Test key prompts monthly in ChatGPT, Claude, Gemini, and Grok
- Use a spreadsheet or database to track visibility score per prompt:

Prompt	ChatGPT	Claude	Gemini	Grok	Mentioned?	Notes
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- Prioritize prompts that align with user purchase or trust moments

2. Content & Platform Audits

- Check freshness of:
 - Medium articles
 - Substack issues
 - Hugging Face READMEs
 - GitHub metadata files
- Validate page indexing with Bing and Google Search Console

3. Metadata & Schema Health

- Run structured data validators:
 - Google Rich Results Test
 - Schema.org validator
- Check for changes to schema standards and update as needed

4. External Mentions & Backlinks

- Monitor mentions with:
 - Brand Alerts (e.g., Google Alerts, Mention.com)
 - SEO tools (Ahrefs, SEMrush)
- Look for increases/decreases in trusted citations

5. Cross-LLM Metadata Parity

Audit that your brand description, categories, and metadata fields match across GPT-5, Gemini, Claude, and other AI ecosystems to maximise cross-model recommendation flow.

13.4 Suggested Visibility Cadence

Task	Frequency
Prompt testing in 4 LLMs	Monthly
Schema and site audit	Quarterly
Content refresh or addition	Monthly
Trust signal re-verification	Semi-annually
Indexing status review	Bi-monthly
Ecosystem profile updates	Quarterly

13.5 Tools & Techniques

- Use prompt-monitoring tools like PromptLayer, PromptMonitor.io
- Create AI assistant personas to test conversational results
- Track changes using website diff checkers and schema comparators
- Implement internal dashboards for visibility KPIs

13.6 Certification Requirements

For continued AIVO Standard™ compliance:

- Maintain prompt visibility logs for 3+ key prompts per quarter
- Refresh or audit key visibility components every 90 days
- Document updates made and their rationale (version control or changelog)
- Maintain redundancy across multiple content ecosystems

13.7 Risks and Mitigation

Risk: Loss of visibility after LLM updates

- Mitigation: Pre-schedule prompt testing and re-submission cycles

Risk: Fatigue or inconsistency in monitoring cadence

- Mitigation: Assign ownership, automate alerts, and maintain centralized checklists

Risk: Platform changes or deprecated standards

- Mitigation: Monitor schema.org, Wikidata, Hugging Face, and OpenAI updates monthly

13.8 Strategic Significance

This final stage ensures that all prior investments continue to generate visibility returns. Monitoring transforms AIVO from a static playbook into a living system, adapting to algorithm shifts, emerging prompt patterns, and evolving AI discovery behavior.

Ongoing visibility work ensures:

- Resilience against search and prompt volatility
- Continuous improvement based on real-world performance
- Competitive advantage in an accelerating AI-first world

14. Quantification Layer: Prompt-Space Occupancy Score (PSOS™)

14.1 Rationale and Purpose

While the preceding stages of the AIVO Standard define the practices necessary to establish and optimize brand visibility across AI ecosystems, a governing Key Performance Indicator (KPI) is required to ensure that visibility can be measured, benchmarked, and audited.

The Prompt-Space Occupancy Score (PSOS™) is introduced as this governing KPI. Analogous to how PageRank crystallized the early web's ranking dynamics into a communicable metric, PSOS™ condenses the volatility and multi-dimensionality of AI visibility into a single, board-ready score.

PSOS™ allows organizations to:

- Establish baselines of AI visibility.
- Monitor longitudinal changes over time.
- Compare performance against peers and industry benchmarks.
- Quantify reputational and resilience factors beyond simple presence.
- Translate technical fluctuations in AI search into strategic risk and ROI discussions at Board level.

14.2 Conceptual Framework

PSOS™ is founded on the principle that AI visibility cannot be reduced to a single rank position, as in traditional search. Instead, it must reflect:

1. Breadth – coverage across a validated set of buyer-relevant prompts.
2. Depth – persistence and salience of appearances over time.
3. Resilience – cross-engine stability, weighted by market share and regional adoption.
4. Sentiment – quality and polarity of how the brand is represented.
5. Decay – rate of decline or persistence across intervals.
6. (SME/Local mode only) Offline Credibility and Confidence Index – to mitigate sparse data bias and ensure trustworthiness.

14.3 Dual Measurement Modes

PSOS™ is deployed in two complementary modes, ensuring applicability across both enterprise and SME/local contexts.

14.3.1 Enterprise Mode (Prompt-Validated PSOS)

- **Methodology:** Direct testing of curated prompt clusters (30–75 prompts; default 50).
- **Execution:** Each prompt is run across multiple AI engines (ChatGPT, Gemini, Claude, Perplexity, Grok), with engine weightings updated annually to reflect market share.
- **Replicates:** Multiple runs per prompt are executed to account for non-determinism, with 95% confidence intervals reported.
- **Scoring Formula:**
 - $PSOS_{Enterprise} = (Breadth \times Depth \times Resilience) + SentimentOverlay - DecayAdjustment$
- **Governance:**
 - Prompt clusters must be versioned and frozen (v1.0, v1.1, etc.) for reproducibility.
 - Quarterly cadence is recommended.
 - Outputs must display formula version, cluster version, and engine-weight provenance to ensure defensibility.

14.3.2 Local/SME Mode (Evidence-Based PSOS)

- **Methodology:** Proxy scoring where direct prompt testing is infeasible.
- **Inputs:** Diversity of citations/reviews (Breadth), structured content depth, platform resilience, review sentiment, freshness of updates, offline credibility (tenure, certifications, dispute resolution).
- **Confidence Index:** A secondary score that qualifies reliability by factoring review volume, platform distribution, variance, and recency.
- **Scoring Formula (weighted composite):**

$$PSOS_{SME} = 0.25B + 0.20D + 0.20R + 0.15S + 0.10F + 0.15O$$
$$PSOS_{SME} = 0.25B + 0.20D + 0.20R + 0.15S + 0.10F + 0.15O$$

- **Interpretation Bands:**
 - Fragile <40
 - Moderate 40–69
 - Strong ≥70

14.4 Attribution and ROI Integration

A defining feature of PSOS™ is its planned integration with causal attribution modeling, enabling organizations to correlate improvements or declines in visibility with measurable business outcomes (leads, opportunities, revenue).

- Methodology: Difference-in-Differences (DiD) and uplift models applied around intervention windows (e.g., new schema deployment, review campaigns).
- Purpose: Elevates PSOS™ from a monitoring metric to a financial KPI that boards can act upon.

14.5 Governance and Auditability

To ensure PSOS™ remains auditable and defensible:

- Version Control: All formulas and prompt sets are locked by version number.
- Provenance Logging: Changes to engine weights, formulas, and cluster versions must include notes, timestamps, and actor attribution.
- Audit Trail: Quarterly or semi-annual exports must include raw evidence snapshots, error bands, refusal/neutral classifications, and weight-mix explanations.
- Board-Ready Reporting: Outputs are structured as one-page executive dashboards, highlighting current score, confidence intervals, drivers of change, competitor displacement, and ROI linkage.

14.6 Position in the AIVO Standard

PSOS™ functions as the Quantification Layer of the AIVO Standard:

- The nine preceding stages define how visibility is built and maintained.
- PSOS™ defines how visibility is measured, validated, and governed.

Together, they establish AIVO not only as a framework of best practices but as a measurable, auditable standard — bridging the gap between operational execution and board-level accountability.

15 — Entropy & Stability

Purpose

This section introduces the Entropy and Stability extensions to the AIVO Standard™, enhancing the existing PSOS framework (Section 14) with quantitative measures of variance, consistency, and governance reliability over time. Where PSOS determines where a brand appears, this layer measures how reliably and predictably that visibility is maintained across prompt variants, models, and audit periods.

These v3.5 extensions formalise three linked metrics:

- PSOS-E (Entropy-Adjusted Occupancy Score) – visibility weighted by stability.
- H (Entropy) – dispersion or uncertainty of brand visibility.
- SI (Stability Index) – resilience of visibility over time.

Where This Fits: Section 14 defines the Prompt-Space Occupancy Score (PSOS). Section 15 builds directly upon it, providing the statistical assurance layer referenced in Stage 9 (Monitoring & Continuous Assurance).

15.1 Key Definitions and Equations

Symbol	Definition	Formula / Description
p_i	Probability of brand appearance in prompt variant <i>i</i>	count of positive appearances / total samples
H (Entropy)	Measure of uncertainty or volatility in brand visibility	$H = - \sum (p_i \times \log_2 p_i)$
H_n (Normalised Entropy)	Entropy scaled 0–1 for comparability	$H_n = H / \log_2 n$
SI (Stability Index)	$1 - H_n$; captures reliability (1 = stable, 0 = chaotic)	$SI = 1 - H_n$
PSOS-E	Entropy-Adjusted Prompt-Space Occupancy Score	$PSOS-E = PSOS \times SI$

Interpretation

- Low H / High SI (> 0.90): visibility is stable and predictable.
- High H / Low SI (< 0.80): volatile visibility – potential collapse risk.
- PSOS-E provides a governance-grade visibility score adjusted for uncertainty.

15.2 Monte Carlo Audit Protocol (MCAP)

To ensure replicable results and statistical confidence, the AIVO Standard employs the Monte Carlo Audit Protocol (MCAP).

Process Steps

- 1.Sampling: ≥ 100 prompt perturbations per intent cluster across ≥ 3 LLM engines.
- 2.Execution: For each sample, record brand presence, rank, and citation tier.
- 3.Entropy Computation: Calculate H and SI per prompt cluster.
- 4.Replication: Re-run with different model seeds and temperatures (0.2 – 0.7); store all run hashes.
- 5.Confidence Interval: Compute 95 % CI for PSOS-E and SI; flag deviations > 0.05 .

Governance Record: Each run must retain model version, temperature, seed, timestamp, and SHA-256 hash for a minimum of 24 months.

15.3 Interpretation and Board Reporting

Control Optimum Zone

Zone	SI Range	Description	Recommended Action
Green Stable	≥ 0.90	Predictable visibility	Maintain cadence and content structure
Amber Watch	0.80 – 0.89	Emerging variability	Increase sampling; review prompt diversity
Red Drift	< 0.80	High volatility / collapse risk	Trigger Continuous Assurance Review (Stage 9)

Board KPI Template

Metric	Definition	Current Value	Target	Status	Trend
PSOS (Base)	Raw Prompt-Space	0.74	0.8	▲	Stable
SI	Stability Index	0.86	≥ 0.90	⚠	Slight decline
ΔH	Entropy change over 30 days	0.04	≤ 0.02	!	Increasing variance
PSOS-E	Entropy-Adjusted	0.64	≥ 0.72	⚠	Requires intervention

Where This Fits: Insert into Stage 9 (Monitoring) dashboard as the “Statistical Assurance Layer.” Triggers alert when SI drops below threshold or ΔH exceeds control band.

15.4 Continuous Assurance Integration

The Entropy & Stability layer feeds directly into the Continuous Assurance Loop (Section 9.0).

Cycle: Audit → Monitor → Alert → Verify

Step	Description	Owner
Audit	Quarterly Monte Carlo runs to refresh H & SI baselines.	Data Ops / AIVO Analyst
Monitor	Monthly entropy trend analysis (+ rolling ΔH).	Continuous Assurance Unit
Alert	SI < 0.8 or ΔH > 0.05 triggers Stage 9 Corrective Action.	Governance Lead
Verify	Confirm remediation and update board scorecard.	AIVO Audit Committee

15.5 Certification Crosswalk

Input	Calculation	Evidence Artifact	Pass/Fail Criteria
Prompt samples (≥ 100 / cluster)	PSOS, H, SI	Audit CSV + Run Hashes	Coverage $\geq 95\%$ confidence
Model variance tests	$\Delta H / \Delta SI$	MCAP Report	$\Delta \leq 0.05$
Board report summary	PSOS-E dashboard	PDF + Sign-off Record	All thresholds within control bands

15.6 Glossary Additions

Term	Definition
PSOS-E	Entropy-Adjusted Prompt-Space Occupancy Score = PSOS $\times$ SI
H (Entropy)	Statistical uncertainty of brand visibility distribution
SI (Stability Index)	Normalised measure of visibility resilience over time (1 = fully stable)
MCAP	Monte Carlo Audit Protocol for replicable sampling and validation
Control Optimum Zone	Target band of SI values indicating acceptable stability (≥ 0.90)

15.7 Summary

Section 15 extends the AIVO Standard into statistical assurance.

By integrating Entropy and Stability into the core PSOS measurement, boards gain quantifiable control over visibility volatility.

This section links directly to:

- Section 14 (PSOS) – provides base inputs for PSOS-E.
- Stage 9 (Monitoring) – defines alert and assurance thresholds.

Together, they ensure AIVO remains the only governance-grade standard capable of auditing both the *existence* and *stability* of AI-era visibility.

15.8 Annex 15-T : Technical Specification Reference

All equations, sampling rules, and computational procedures defined in this section are governed by the AIVO Standard Technical Specification – Entropy & Stability (v3.5), hereafter referred to as Annex 15-T.

Annex 15-T forms the authoritative statistical reference for:

- Formal mathematical definitions of PSOS-E, Entropy (H), and Stability Index (SI);
- The full Monte Carlo Audit Protocol (MCAP) covering sampling size, model temperature ranges, and replication logic;
- Provenance, replication-hash, and data-retention standards;
- Validation thresholds and ISO 42001 / NIST AI RMF compliance mapping;
- Worked examples and sectoral baseline tables.

Methodology readers should consult **Annex 15-T** for implementation-level requirements, numerical validation, and code-level reproducibility.

The present section defines *governance and measurement intent*; Annex 15-T defines *execution precision and conformance tests*.

Methodological Process

Purpose

This section introduces the Two-Dimensional Prompt Measurement Model, an expansion of the Prompt-Space Occupancy Score (PSOS) that separates visibility into two interpretable layers of interaction within Large Language Models (LLMs): Awareness and Transaction.

Where the original PSOS quantifies if a brand appears, this framework quantifies how and when it appears in the user’s decision-making dialogue. The model provides boards, CMOs, and data-governance teams with a dual-axis diagnostic for measuring both brand discovery and conversion pathway inclusion inside generative ecosystems.

Key Definitions

Metric	Description	Example
PSOS-A (Awareness Layer)	Measures presence in <i>initial discovery or evaluative prompts</i> (e.g., “best travel site for booking hotels”). It reflects the brand’s inclusion in first-layer assistant recommendations.	Expedia appears in the top five results of a general travel query.
PSOS-T (Transaction Layer)	Measures presence in <i>follow-up transactional prompts</i> (e.g., “where can I book this hotel?”). It quantifies conversion-path exposure—the ability of the assistant to surface the brand as the execution channel.	Booking.com returned when a user asks where to complete a booking.
PSOS-E (Entropy-Adjusted Composite)	Weighted measure of persistence and reliability across both layers, integrating the Stability Index (SI) from Section 15.	Brand maintains consistent presence across dialogue turns despite model drift.

Methodological Process

1. Dialogue Decomposition

Identify the most common intent pathways for a given sector.

Each pathway comprises an **Awareness Prompt (AP)** (discovery-oriented) and at least one **Transactional Prompt (TP)** (action-oriented).

Example for Travel Sector:

AP = "Best family resorts in Spain"

TP = "Where can I book the best family resorts in Spain?"

2. Dual-Phase Sampling

Each AP/TP pair is run across ≥ 3 LLMs (GPT-4, Gemini 2.5, Claude 3 Opus, Perplexity Pro).

For each phase:

- Record brand appearance, rank, and citation tier.
- Compute independent PSOS-A and PSOS-T values.
- Apply Monte Carlo Audit Protocol (MCAP) from Annex 15-T for variance validation.

3. Cross-Layer Linkage

Map Awareness to Transaction using session thread matching (LLM conversation persistence ID).

This identifies whether brands recommended initially are also retained in follow-up prompts.

4. Computation Model

$$\text{PSOS}_2\text{D} = (\text{PSOSA} + \text{PSOST}) / 2$$

$$\text{PSOSE} = \text{PSOS}_2\text{D} \times \text{SI}$$

Where SI = Stability Index (from Section 15).



5. Interpretation

Range	Interpretation	Governance Action
≥ 0.75	Strong multi-layer visibility	Maintain cadence & monitor entropy.
0.50 – 0.74	Uneven layer performance	Target prompt clusters with low transactional retention.
< 0.50	Fragmented or volatile presence	Escalate to PSOS Risk Probe for root-cause analysis.

Board-Level Application

Visibility Governance Metric. Boards track the 2D PSOS score to quantify conversion-path representation within AI assistants, equivalent to share-of-search in traditional SEO.

Revenue Assurance. Correlation between PSOS-A and PSOS-T indicates pipeline retention across AI-mediated journeys. Low PSOS-T exposes leakage between awareness and purchase.

Operational Benchmarking. Quarterly 2D PSOS scores by category support budget re-allocation to prompts or knowledge sources with highest conversion leverage.

Risk Flagging. A discrepancy > 0.25 between Awareness and Transaction scores triggers an AI Visibility Governance alert (Stage 9 Monitoring).

Governance & Reporting

- **Sampling Integrity:** All runs must retain prompt logs and hashes in line with MCAP (Annex 15-T).
- **Audit Frequency:** At least bi-annual per category; quarterly for regulated sectors (Financial Services, Healthcare).
- **Assurance Review:** Scores formally reported in the AIVO Board Audit Pack under the Visibility Assurance KPI group.
- **Continuous Improvement:** Where variance > 10 % between PSOS-A and PSOS-T, an adaptive content and citation review is mandated.

Cross-References

- **Section 14:** Prompt-Space Occupancy Score (PSOS) — Core Metric
- **Section 15:** Entropy & Stability Extensions — SI Computation
- **Annex 15-T:** Monte Carlo Audit Protocol (MCAP)
- **Section 17 (forthcoming):** Predictive Conversational Models (PCM) — Temporal Prompt Path Simulation

Summary for Boards

The Two-Dimensional Prompt Measurement Framework extends the AIVO Standard from static visibility to dialogue-path visibility. It quantifies a brand's ability to survive and convert across the two critical layers of LLM interaction: awareness and transaction. The resulting 2D-PSOS score provides boards with a governance-grade indicator of AI conversion resilience and revenue protection within assistant-driven markets.

17 Predictive Conversational Models (PCM)

Purpose

This section introduces Predictive Conversational Modelling (PCM) — a framework for forecasting how LLM-driven dialogues evolve over multiple user turns, and where brands can position themselves for discovery and transaction opportunities.

While PSOS quantifies presence, and the 2D-PSOS framework measures layered retention, PCM predicts future conversational states that lead to commercial or reputational outcomes.

The methodology formalises a repeatable process for modelling user intent, predicting dialogue trajectories, and identifying the most probable prompt sequences that drive visibility, conversion, or substitution.

Core Definition

Predictive Conversational Models (PCM) simulate the probabilistic flow of dialogue between users and LLMs using structured prompt pathways, intent classification, and entropy-adjusted transition probabilities.

Each PCM identifies:

1. **Intent Nodes** — Represent distinct user goals (awareness, comparison, transaction, post-purchase).
2. **Transition Probabilities (P_t)** — Likelihood of progressing from one node to another.
3. **Visibility State (V_s)** — Probability that a brand is surfaced or retained during each transition.

Methodological Framework

1. Dialogue Mapping (Intent Topology)

Define the canonical conversational graph for a given category.

Example for Travel Sector:

Node 1: "I want a cheap hotel in Amsterdam" → Awareness

Node 2: "Can you narrow it down near the museums?" → Refinement

Node 3: "Which ones include breakfast?" → Preference Filter

Node 4: "Where can I book this?" → Transaction

Each transition (1→2, 2→3, etc.) has a measurable probability and associated brand visibility score.

2. Transition Probability Calculation

For each transition T_{ij} between nodes i and j :

$Pt(i,j) = \text{Frequency of transition } i \rightarrow j / \text{Total transitions from } i$

When combined with brand visibility $Vs(i)$, the expected visibility retention at node j is:

$$Vs(j) = Vs(i) \times Pt(i,j)$$

This allows cumulative simulation of visibility decay or amplification along conversational chains.

3. Monte Carlo Conversation Simulation (MCCS)

Run $\geq 1,000$ simulated dialogues per sector using stochastic intent sampling and prompt perturbation.

For each run:

- Randomise initial awareness prompt (as per Section 16).
- Apply transition probabilities PtP_tPt with entropy noise from Section 15.
- Track visibility states VsV_sVs at each conversational depth.
- Output metrics:
 - **Conversation Depth Resilience (CDR)** – mean number of turns before brand omission.
 - **Visibility Retention Ratio (VRR)** – percentage of simulations where the brand persists through ≥ 3 dialogue turns.



4. Predictive Visibility Index (PVI)

Derived from the Monte Carlo outputs:

$$PVI = \Sigma(Vs(i) \times Pt(i)) / n$$

Interpretation:

Range	Interpretation	Governance Implication
≥ 0.75	Predictable, high-visibility conversation path	Maintain cadence; candidate for long-term model seeding.
0.50–0.74	Moderate predictability with entropy drift	Review content and entity consistency.
< 0.50	Volatile or substitution-prone visibility	Trigger PSOS Risk Probe and citation reinforcement plan.

5. Conversational Outcome Mapping

Link predicted conversational states to measurable business outcomes:

- Discovery → Conversion: identifies how often awareness prompts transition into booking/transaction prompts.
- Conversion → Retention: assesses whether brand continues to appear in service/support-related queries post-purchase.
- Deflection: models pathways where users divert to competitors or misinformation sources.



Governance & Board-Level Use

Governance Lens	Description
Visibility Risk Forecasting	PCM allows boards to anticipate visibility collapse by modelling high-entropy conversational paths.
Revenue Protection	By linking predicted conversion paths to PSOS-T, boards quantify <i>Revenue at Conversational Risk (RaCR)</i> .
Operational Intelligence	Marketing, SEO, and AI Governance teams gain early warning of emerging conversational biases or topic drifts.
Regulatory Readiness	PCM supports ISO 42001 §8.2 and EU AI Act Article 55 compliance through auditable explainability of AI-influenced discovery processes.

Integration with AIVO Ecosystem

- **Feeds:** PSOS (Section 14) → 2D-PSOS (Section 16) → PCM (Section 17).
- **Dependencies:** Entropy & Stability (Section 15) provides the noise model used in Monte Carlo simulation.
- **Outputs:** Predictive Visibility Index (PVI) feeds into AIVO Risk Dashboard and Board Audit Packs.

Board Summary

Predictive Conversational Models extend the AIVO Standard from reactive measurement to proactive governance. By simulating multi-turn dialogue pathways, organisations can forecast visibility decay, anticipate revenue leakage, and position themselves within conversational flows that define the AI-driven economy.

Annex 15-T — Technical Specification: Entropy, Stability & MCAP

Document Type: AIVO Standard Technical Annex

Version: v3.5

Linked Sections: Methodology Section 15 (Entropy & Stability Extensions)

Status: Approved for Publication under AIVO Standard™ Methodology Version 3.5

15-T.1 Purpose & Scope

This annex defines the formal mathematical, computational, and governance framework that operationalises the *Entropy & Stability Extensions* described in Section 15 of the AIVO Standard™.

It serves as the authoritative specification for implementing PSOS-E (Entropy-Adjusted Prompt-Space Occupancy Score), H (Entropy), and SI (Stability Index) through a validated Monte Carlo Audit Protocol (MCAP).

Objectives

- Establish a replicable quantitative standard for measuring visibility stability within non-deterministic LLM environments.
- Define sampling, calculation, and provenance rules to ensure audit reproducibility.
- Map conformance requirements to ISO 42001, EU AI Act, and NIST AI RMF 1.0.

15-T.2 Mathematical Definitions and Derivations

15-T.2.1 Entropy (H)

For a discrete distribution of brand appearance probabilities

$$H = - \sum (p_i \times \log_2(p_i))$$

15-T.2.2 Normalised Entropy (H_n)

$$H_n = H / \log_2(n)$$

Ranges 0–1, where 0 = perfect predictability and 1 = complete disorder.

15-T.2.3 Stability Index (SI)

$$SI = 1 - H_n$$

Interpreted as the proportion of predictable variance in visibility.

15-T.2.4 Entropy-Adjusted PSOS (PSOS-E)

$PSOSE = PSOS \times SI$

Weights baseline occupancy by stability to produce a **governance-grade visibility metric**.

Prompt Cluster	n	p _i (avg)	H	SI	PSOS	PSOS-E
Hotels – Amsterdam	100	0.42	0.79	0.86	0.74	0.64

15-T.3 Monte Carlo Audit Protocol (MCAP)

15-T.3.1 Sampling Requirements

- ≥ 100 prompt perturbations per intent cluster.
- ≥ 3 LLM models (OpenAI GPT-4, Gemini 2.5 Flash, Claude 3, etc.).
- Randomised temperature (0.2 – 0.7) and seed per run.

15-T.3.2 Execution Process

- 1.Generate prompt variants using controlled synonym swaps and order permutations.
- 2.Run each variant across engines; record brand appearance, rank, and citation tier.
- 3.Compute H and SI for each cluster.
- 4.Repeat 5 times with unique seeds; store hash of each run.
- 5.Calculate mean PSOS-E and 95 % confidence interval.

15-T.3.3 Validation Criteria

Metric	Target	Tolerance	Action
ΔH (between runs)	≤ 0.05	± 0.02	Re-sample if exceeded
ΔSI	≤ 0.05	± 0.03	Flag volatility
Coverage	≥ 95 %	–	Extend sampling if below

15-T.4 Provenance & Integrity Requirements

Metadata Per Run

- Model ID and version
- Temperature and seed
- Timestamp and prompt set ID
- SHA-256 hash of input/output
- Replication Key (JSON)

Retention

All audit records must be retained for ≥ 24 months in encrypted storage (AES-256).

Reproducibility Test

Independent auditors must be able to recreate results within $\pm 3\%$ of original PSOS-E values using the Replication Key.

15-T.5 Governance & Certification

15-T.5.1 Conformance Levels

Level	Description	Evidence Requirements
Level I – Compliance	Meets minimum sampling + record standards	MCAP Report, Run Hashes
Level II – Assurance	Validated across ≥ 3 models + cross-seed replicates	Variance Report (ΔH , ΔSI)
Level III – Governance	Full Continuous Assurance Cycle integrated with board KPI review	Board Dashboard + Sign-off PDF

15-T.5.2 Regulatory Alignment

Framework	Clause	Alignment Summary
ISO 42001 (AI Management System)	§ 7.4, 8.2	Continuous assurance of AI outputs via measurable metrics
EU AI Act (Draft 2025)	Art. 52–55	Transparency & traceability of high-risk systems
NIST AI RMF 1.0	Map / Measure / Manage / Govern	Metric-based risk governance
OECD AI Principles	#2, #3	Transparency and accountability via replicable audits

15-T.6 Bias & Modality Assessment

- If $H > 0.4$ in any language subset, flag for bias review.
- Compute SI per language and region; deviation > 0.1 = risk indicator.
- For multimodal systems, test text vs image/audio equivalence by normalising response entropy.

15-T.7 Sectoral Baselines & Examples

Sector	Mean PSOS	Mean SI	PSOS-E	Risk Band
Travel & Hospitality	0.74	0.86	0.64	Medium
Retail E-Commerce	0.7	0.9	0.63	Low
Healthcare	0.68	0.82	0.56	High
Financial Services	0.72	0.88	0.63	Medium

Entropy > 0.15 above sector mean should trigger re-audit under Stage 9.

15-T.8 Appendices

A. Sample Entropy/SI Computation (Python Pseudocode)

```
import math
def entropy(p):
    return -sum([x*math.log2(x) for x in p if x>0])
def stability_index(p):
    h = entropy(p)
    hn = h / math.log2(len(p))
    return 1 - hn
def psos_e(psos, p):
    return psos * stability_index(p)
```

B. Replication Key (JSON Schema)

```
{
  "run_id": "UUID",
  "model_id": "gpt-4",
  "temperature": 0.4,
  "seed": 12345,
  "timestamp": "2025-10-06T12:00:00Z",
  "hash": "SHA256_STRING"
}
```

C. References & DOIs

- AIVO Standard Methodology v3.5 – Zenodo DOI [placeholder].
- ISO 42001 AI Management System Standard (2023 draft).
- NIST AI Risk Management Framework 1.0 (2023).
- EU AI Act Consolidated Text (2025).
- AIVO Standard Technical Dataset v3.5 (Entropy & Stability) – Zenodo DOI [placeholder].

15-T.9 Version Control

Version	Date	Author	Description
1	10/6/2025	AIVO Standards Board	Initial technical annex based on AIVO 3.5 specification

Annex 17-T — Technical Specification: Predictive Conversational Models (PCM)

Document Type: AIVO Standard Technical Annex

Version: v3.5

Linked Sections: Section 17 (Predictive Conversational Models) and Section 16 (2D-PSOS)

Status: Approved for Publication under AIVO Standard™ Methodology v3.5

17-T.1 Purpose & Scope

This annex defines the mathematical, computational, and governance framework for Predictive Conversational Models (PCM) within the AIVO Standard™.

It provides implementers with the algorithms, parameters, and audit structures required to simulate multi-turn LLM dialogues and compute predictive visibility metrics.

Objectives

- Establish reproducible conversation-path simulations linking Awareness (PSOS-A) and Transaction (PSOS-T) layers.
- Quantify brand persistence through stochastic multi-turn transitions.
- Map predictive outcomes to governance metrics such as Visibility Retention Ratio (VRR) and Predictive Visibility Index (PVI).
- Align with ISO 42001 § 8.2, EU AI Act Articles 52–55, and NIST AI RMF 1.0 for transparency and continuous assurance.

17-T.2 Mathematical Definitions and Core Equations

17-T.2.1 Transition Probability P_t

$$P_t(i,j) = \text{Frequency of transition } i \rightarrow j / \text{Total transitions from } i$$

Represents the likelihood that a conversation moves from intent node i to j .

17-T.2.2 Visibility State V_s

$$V_s(j) = V_s(i) \times P_t(i,j)$$

Denotes the probability that a brand remains visible when a transition occurs.

17-T.2.3 Conversation Depth Resilience (CDR)

$$\text{CDR} = (1 / n) \times \sum(dk)$$

Average number of dialogue turns before brand omission across n simulations.

17-T.2.4 Visibility Retention Ratio (VRR)

$$\text{VRR} = \text{Simulations with } \geq 3 \text{ visible turns} / \text{Total simulations}$$

17-T.2.5 Predictive Visibility Index (PVI)

$$\text{PVI} = \sum(Vs(i) \times Pt(i)) / n$$

Entropy-adjusted mean of expected visibility across all transitions.

17-T.3 Monte Carlo Conversation Simulation (MCCS)

Sampling Parameters

- $\geq 1\,000$ conversation runs per sector.
- Randomised initial prompt sets (≥ 100 variants).
- 3–4 LLM engines (GPT-4, Gemini 2.5, Claude 3, Perplexity Pro).
- Temperature range 0.2 – 0.7 with ≥ 5 unique seeds per engine.
- Entropy noise applied from Section 15 (Entropy & Stability).

Execution Process

1. Generate intent graph (nodes = awareness $\rightarrow$ transaction $\rightarrow$ post-transaction).
2. For each node pair (i, j) , record transition frequency and brand visibility.
3. Compute $Pt(i, j)$ and $Vs(j)$.
4. Iterate 1 000 times using Monte Carlo sampling to produce distribution of PVI values.
5. Store hashes and replication keys for each run (see § 17-T.6).

Validation Criteria

Metric	Target	Tolerance	Action
Δ PVI (between runs)	≤ 0.05	± 0.02	Re-sample if exceeded
VRR	≥ 0.60	–	Flag if below
CDR	≥ 3 turns	± 1	Review conversation depth
Coverage	$\geq 95\%$	–	Extend sampling if below

17–T.4 Data Structures and JSON Schema

Conversation Node Schema

```
{
  "node_id": "UUID",
  "intent_type": "awareness|transaction|support",
  "prompt_text": "string",
  "transitions": [
    { "to": "node_id", "p_t": 0.42, "v_s": 0.65 }
  ],
  "model_id": "gpt-4",
  "temperature": 0.4,
  "seed": 12345,
  "timestamp": "2025-10-06T12:00:00Z"
}
```

Replication Key Schema

```
{
  "simulation_id": "UUID",
  "engine": "gemini-2.5-flash",
  "entropy_source": "Annex15-T",
  "conversation_depth": 4,
  "hash": "SHA256_STRING"
}
```

17-T.5 Governance and Certification

Level	Description	Evidence Requirements
I – Compliance	Basic PCM simulation completed with replication key storage	Run log + PVI distribution
II – Assurance	Validated across ≥ 3 models with cross-seed replicates	Monte Carlo variance report (Δ PVI, VRR)
III – Governance	Continuous Assurance integrated with AIVO Risk Dashboard	PCM Audit Dashboard + Board Sign-off

Regulatory Alignment

Framework	Clause	Alignment Summary
ISO 42001 (AI MS)	§ 7.4 / 8.2	Predictive monitoring of AI outputs via quantified dialogue modelling
EU AI Act (2025 Draft)	Art. 52–55	Traceability and auditable simulation of AI recommendation paths
NIST AI RMF 1.0	Map / Measure / Manage / Govern	Integrates risk measurement into AI governance loop
OECD AI Principles	#2, #3	Transparency and accountability through predictive simulation

17-T.6 Provenance and Integrity Requirements

- Each simulation run must include: engine version, prompt set ID, timestamp, temperature, seed, and hash.
- SHA-256 hash chains for replication integrity.
- Retention ≥ 24 months in AES-256 encrypted archive.
- Independent auditors must reproduce PVI within $\pm 3\%$ variance.

17-T.7 Sectoral Baselines and Benchmarks

Sector	Mean PVI	VRR	Risk Band
Travel & Hospitality	0.74	0.68	Medium
Retail E-Commerce	0.77	0.72	Low
Healthcare	0.65	0.55	High
Financial Services	0.73	0.67	Medium

A deviation > 0.10 below sector mean requires Stage 9 Governance review.

17-T.8 Appendices

A. Example Transition Matrix

From $\rightarrow$ To	P_t	V_s	Description
Awareness $\rightarrow$ Refinement	0.82	0.9	Brand retained in second query
Refinement $\rightarrow$ Transaction	0.67	0.79	Partial drop in visibility
Transaction $\rightarrow$ Post-Purchase	0.54	0.71	Brand remains visible in support query



B. Python Pseudocode — PCM Simulation

```
import random, math

def simulate_dialogue(nodes, transitions, runs=1000):
    results = []
    for _ in range(runs):
        v_s = 1.0
        for (i, j, p_t) in transitions:
            if random.random() < p_t:
                v_s *= p_t
        results.append(v_s)
    pvi = sum(results) / len(results)
    return pvi
```

C. References & DOIs

- AIVO Standard Methodology v3.5 — Zenodo DOI [placeholder]
- ISO 42001 AI Management System Standard (2023 draft)
- NIST AI Risk Management Framework 1.0 (2023)
- EU AI Act Consolidated Text (2025)
- AIVO PCM Technical Dataset v3.5 — Zenodo DOI [placeholder]

17–T.9 Version Control

Version	Date	Author	Description
1	10/6/2025	AIVO Standards Board	Initial technical annex based on AIVO 3.5 PCM specification

Appendix A – Tools, Templates & Validators

This appendix provides ready-to-use links, validators, and templates to simplify implementation of each AIVO stage.

Recommended Tools by Stage

Stage	Tool	Description	Link
2	Schema Markup Generator	Generate JSON-LD for Organization, Product, FAQ, etc.	Web tool
3	Crunchbase	Add company profile for mention graph credibility	Public directory
4	PromptMonitor.io	Track and test prompt visibility across LLMs	Visibility monitor
5	Medium / Substack	Publish trusted, AI-indexed articles	Publishing platforms
6	Bing Webmaster Tools	Submit your URLs for crawling	Indexing tool
6	Perplexity Submission Form	Submit your product URL to Perplexity	Discovery form
7	Hugging Face Spaces	Create LLM-indexed product tools or demos	AI ecosystem hub
7	Custom GPT Creator	Build assistant tied to your business	OpenAI tool
9	Google Rich Results Test	Validate structured data on your site	Validator

Templates

- Prompt testing tracker (CSV/Google Sheet)
- Visibility audit checklist
- Metadata consistency reference sheet
- Trust signal capture sheet
- Quarterly monitoring log

(Templates can be hosted as downloadable assets or GitHub links)

Certification Readiness Checklist

Stage	Evidence Required	Verification Method	Submitted (Y/N)
1	Target prompts list + goal alignment document	Strategy log or discovery prompt map	
2	Live schema, Wikidata, org metadata	Screenshots + schema validation report	
3	Listings in Crunchbase, GitHub, etc.	Public URLs + profile screenshots	
4	Prompt examples + website usage guide	Web capture of prompt anchors	
5	Published content links with metadata	URLs + LLM prompt test responses	
6	Submissions to Bing, Perplexity, etc.	Submission receipts or indexing tools	
7	Custom GPT / Hugging Face profile live	Screenshots + assistant link	
8	Trust signals (testimonials, cross-links)	Live references + backlink audits	
9	Monitoring dashboard or logs	Prompt log spreadsheet + audit record	

Appendix B – Use Case Scenarios

1. AI Therapy Companion

- **Objective:** Be recommended for prompts like “AI support for anxiety”
- **AI VO Tactics:**
 - Schema markup with mental health tags
 - Publishing on Medium/Substack with CBT-related metadata
 - Wikidata entity linking to anxiety, CBT, and AI support
 - Custom GPT trained to role-play emotional support

2. SaaS Productivity Tool

- **Objective:** Surface in queries like “Best AI tools for time management”
- **AI VO Tactics:**
 - GitHub README with clear use cases and YAML metadata
 - Testimonials on G2/Trustpilot
 - Prompt testing across 4 LLMs monthly
 - Submissions to Bing and Perplexity for indexing

3. EcoTech or Climate Startup

- **Objective:** Appear for “sustainability tools” or “carbon offsetting startups”
- **AI VO Tactics:**
 - Crunchbase and Wikidata entries with environmental metadata
 - Mentions in green-tech blogs and Substack posts
 - Cross-link with grant directories and government climate resources

4. Consulting Business or Coach

- **Objective:** Be discoverable under prompts like “AI business coach”
- **AI VO Tactics:**
 - Create a Custom GPT based on coaching framework
 - Publish frameworks as Hugging Face Space or GitHub project
 - Collect client reviews and embed structured testimonials

5. Open Source AI Tool

- **Objective:** Ranked in “alternatives to [popular tool]” prompts
- **AI VO Tactics:**
 - Detailed GitHub repo with LICENSE, issues log, contributions
 - Schema.org Product + SoftwareApplication metadata
 - Tech writers or community advocates linking on Reddit/Substack

Appendix C – Ethical Guidelines & LLM Alignment

AIVO is designed to align with ethical discovery principles and AI safety concerns. Entities implementing this methodology are encouraged to adopt the following guidelines:

✓ **Transparency**

- Disclose when AI is used in user-facing tools
- Provide clear metadata and ownership of content
- Include contact and verification info in schema, Wikidata, and websites

✓ **Authenticity**

- Avoid fabricated reviews or citations
- Ensure all trust signals are verifiable and based on user consent
- Maintain content freshness to reflect actual product capabilities

✓ **Responsible Optimization**

- Do not attempt to “trick” LLMs through keyword stuffing or artificial prompts
- Focus on semantic alignment and public interest, not manipulation

✓ **Accessibility & Inclusion**

- Create ecosystem profiles accessible to a wide range of users
- Use inclusive language and culturally neutral metadata
- Consider ethical implications of visibility (e.g., sensitive topics)

✓ **Alignment with LLM Use Policies**

- Monitor changes to LLMs’ terms of service and data ingestion practices
- Engage in good-faith use of GPTs, Spaces, and Wikidata entities
- Avoid deceptive behavior or false impersonation of competitors

Adaptations for Regulated & High-Trust Industries

While the AIVO Standard™ is designed to be industry-agnostic and universally applicable, certain sectors require enhanced trust, compliance signals, and additional citation depth to be reliably recommended by leading AI assistants such as ChatGPT, Gemini, Claude, and Grok.

This section outlines how to apply the AIVO methodology in **regulated and high-trust environments**, ensuring that businesses in finance, healthcare, insurance, law, and education can benefit from AI Visibility while maintaining credibility, compliance, and trustworthiness.

Why Adaptation is Necessary

Large Language Models (LLMs) are inherently cautious about recommending businesses in sensitive sectors without:

- Verified trust signals
- Transparent data structures
- Third-party credibility
- Strong semantic and factual grounding

Therefore, successful AI Visibility in these industries depends not only on presence but also on proof.

Sectors Requiring Enhanced AIVO Implementation

Sector	Examples
Healthcare	Clinics, therapists, health startups
Finance	Investment platforms, lending, banking
Insurance	Brokers, underwriters, digital insurers
Legal Services	Solicitors, law firms, contract services
Education	Universities, e-learning, certification bodies

These industries typically require additional alignment with local or international regulation (e.g., FCA, SEC, FDA, HIPAA, GDPR, Bar Council), meaning their AIVO implementation must reflect this.

AIVO Pillar	Regulated Sector Enhancement
1. Foundational Presence	Use industry-specific Schema.org types (e.g., MedicalOrganization, FinancialProduct) and include certifications in structured data.
2. Knowledge & Mention Graphs	Submit listings to trusted directories (e.g., FCA register, NHS directory, Bar Council, Crunchbase, G2) and include accreditation references.
3. Prompt Visibility Tools	Use prompt frames that emphasize safety, compliance, and reviews: “FCA-approved brokers UK”, “Top HIPAA-compliant mental health apps”.
4. Publishing & Presence	Publish content on respected sector channels (e.g., PubMed, Lexology, FCA blog, BMC, or sector-specific LinkedIn newsletters).
5. Submission & Indexing	Submit to government or sector-indexed platforms (e.g., gov.uk, clinicaltrials.gov, ISO databases). Verify sitemap accessibility and opt-in status.
6. LLM Profile Building	Custom GPTs, Assistants, or Hugging Face Spaces must include disclaimers (e.g., “This does not constitute legal/medical advice”).



Prompt Framing for High-Trust Visibility

“Trust” prompts require different NLP inputs than typical “best of” queries. Example Prompts for AI Visibility:

- “FCA-regulated life insurance broker in London”
- “Top-rated therapy clinic GDPR compliant”
- “Most trusted ISA investment platforms UK”
- “HIPAA-compliant chronic pain app”

These prompts not only reflect customer intent but align with how LLMs assess confidence and factuality.

Additional Recommendations

- Include references to certifications, professional affiliations, and governing bodies in all metadata, mentions, and custom assistants.
- Use structured publishing on trusted sites to feed LLM memory (e.g., sector-specific Medium blogs, Substack, or industry publications).
- For maximum LLM discoverability, ensure your business is listed on regulatory, accreditation, and peer-review sites.



What's New in AIVO Standard™ v2.1 — GPT-5 Ready

This version introduces enhancements to adapt to GPT-5's expanded retrieval, reasoning, and multi-modal capabilities:

1. Persistent Retrieval Presence — monthly refresh of structured data and citations.
2. Contextual Relevance Across Token Spans — cluster brand mentions across related sources.
3. Citation Fidelity under Ambiguity — use aliases and sameAs linking to avoid confusion.
4. Agentic Integration Potential — presence in APIs and tools GPT-5 calls directly.
5. Source Trust Signaling — prioritise peer-reviewed, government, and academic sources.
6. Multi-Modal Asset Optimisation — captions, alt-text, transcripts for image, video, and audio content.
7. Cross-LLM Metadata Parity — align metadata across GPT-5, Gemini, Claude, and others.



What's New in AIVO Standard™ v2.2 — Visual Search Readiness

This version extends the GPT-5 Ready framework introduced in v2.1 by adding formal guidance for Visual Search & Multi-Modal Asset Readiness — ensuring that brands are optimised for image-based AI discovery channels such as Google Lens, Bing Visual Search, and Pinterest Lens.

Key additions:

- **Stage 5.11 – Visual Search & Multi-Modal Asset Readiness**
 - Comprehensive checklist for optimising images with descriptive filenames, alt text, ImageObject schema, multiple angles, compression, and brand/logo recognition testing.
 - Guidance on maintaining image sitemaps, contextual linking, and AI Overview readiness for visual assets.
 - Prompt visibility testing for visual queries to confirm LLM recognition and indexing.
- **Stage 6 Cross-Reference**
 - Explicit requirement to submit and index optimised visual assets in discovery tools and image search ecosystems.
- **Strategic Integration**
 - Positions visual search as a Tier 1 AI discovery channel feeding structured data directly into AI Overviews, featured snippets, product carousels, and LLM training sets.
 - Emphasises its role in zero-click branding where the image itself is the answer.

With v2.2, the AIVO Standard™ now formally recognises that AI visibility is multi-modal — and that image optimisation is as critical as text for securing recommendation presence in the AI layer.

What's New in Version 3.0 (August 2025)

Version 3.0 of the AIVO Standard™ represents the most significant evolution of the framework since its initial release. While previous versions established a nine-stage methodology for building AI visibility, this edition introduces a formal quantification layer to measure, benchmark, and govern outcomes.

1. Introduction of the Quantification Layer — Prompt-Space Occupancy Score (PSOS™)

- Added as **Section 14.0** following Stage 9.
- Provides a single, board-ready KPI (0–100) that measures how strongly a brand occupies AI recommendation space across large language models (LLMs).
- Includes two complementary modes:
 - **Enterprise Mode (Prompt-Validated PSOS)**: Direct prompt-cluster testing across multiple AI engines, with confidence intervals, refusal/neutral classification, and engine-weight provenance.
 - **Local/SME Mode (Evidence-Based PSOS)**: Proxy measurement using reviews, citations, structured content depth, freshness, offline credibility, and a mandatory Confidence Index.
 -

2. Governance and Auditability Enhancements

- Prompt clusters are now **frozen and versioned** (v1.0, v1.1, etc.) for reproducibility.
- All formulas carry version identifiers (e.g., psos_v1.0.0) to ensure comparability across reporting periods.
- **Confidence intervals** and error bands provide statistical robustness for Enterprise Mode.
- **Refusal/neutral answer classification** highlights white-space opportunities.
- **Provenance logging** for engine weighting ensures transparency when LLM market shares change.



3. Board-Level Reporting Improvements

- Exportable dashboards now include:
 - Headline PSOS with 95% confidence intervals.
 - Sub-indices (Breadth, Depth, Resilience, Sentiment, Decay).
 - “What moved PSOS” analysis identifying drivers of change.
 - Competitor displacement and white-space detection.
- Introduction of a **causal attribution methodology**, enabling organizations to correlate PSOS movements with business outcomes such as leads, opportunities, and revenue.

4. Expanded Appendices and Tools

- Appendices updated to align with PSOS integration, including recommended validators for prompt clusters, review datasets, and schema auditing.
- Tool stack expanded to support both Enterprise and Local/SME measurement modes.

Summary

Version 3.0 transforms the AIVO Standard from a **best-practice methodology** into a **measurable, auditable standard**. By introducing PSOS™, organizations can not only implement visibility strategies but also prove, benchmark, and financially justify them at board level.



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 [AIVO Standard](#)